Thomas C. Fraser

Curriculum Vitae
November 27, 2022
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Education

2018 – Present Ph.D., Quantum Information
Perimeter Institute For Theoretical Physics, Waterloo, ON

2017 – 2018 M.Sc., Theoretical Physics, Quantum Information
Thesis: A Combinatorial Solution to Causal Compatibility
Perimeter Scholars International Program, Perimeter Institute For Theoretical Physics, Waterloo, ON

2012 – 2017 B.Sc., Mathematical Physics, Astrophysics Specialization
University of Waterloo, Waterloo, ON

Academic Works

Papers

2022 *A sufficient family of necessary inequalities for the compatibility of quantum marginals*
Thomas C. Fraser

2020 *Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory*
D. Schmid, T. C. Fraser, R. Kunjwal, A. B. Sainz, E. Wolfe, R. W. Spekkens

2020 *A Combinatorial Solution to Causal Compatibility*
Thomas C. Fraser
Journal of Causal Inference

2018 *Machine Learning Peeling and Loss Modelling of Time-Domain Reflectometry*
J. R. Rinehart, J. H. Béjanin, T. C. Fraser, M. Mariantoni

2018 *Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario*
Thomas C. Fraser, Elie Wolfe
Physical Review A

Talks

2022 The IQC-QuICS Math and Computer Science Seminar
A sufficient family of necessary inequalities for the quantum marginals problem

2021 Beyond IID in Information Theory 9
Simultaneous Spectral Estimation and the Quantum Marginal Problem

2016 Quantum Networks Conference at International Institute for Physics, Natal, Brazil
Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario

TEACHING

WINTER 2021	Quantum Information - Perimeter Scholars International (PSI) Program *Teaching Assistant*
WINTER 2021	Which causal structures admit quantum classical gaps? - PSI Winter School *Student project co-lead*
FALL 2018	Physics 1 - Phys 111 - University of Waterloo *Teaching Assistant*

AWARDS & SCHOLARSHIPS

2022	Natural Sciences and Engineering Research Councilof Canada (NSERC) *Three years, Grant No. 411301803*
2018	UW Graduate Award & Marie Curie Award *Sufficiently High Entrance Average, Department Selected*
2018	The Joanne Cuthbertson and Charlie Fischer Graduate Student Award *Support for an exceptional Masters student from the 2017/2018 PSI class*
2017	Governor General's Silver Academic Medallion *Highest Graduating Average in Graduating Class*
2016	Mike Lazaridis Scholarship *Theoretical Physics Fellowship at Perimeter Institute*
2015	Xerox Research Centre of Canada Limited Award *Best Work-term Report "Acoustic Modelling Using Mel-Frequency Cepstral Coefficients"*
2015	C. C. Lim Physics Prize *Top Marks in Undergraduate Thermodynamics*
2013	Don E. Brodie Scholarship *Highest Experimental Physics Lab Performance*
2012	A. Donald Maynes Scholarship *Outstanding Academic Record*
2012	BMO Undergraduate Entrance Scholarship *Outstanding Academic Average*

INDUSTRY EXPERIENCE

Research & Development Data Scientist

SYSOMOS. TORONTO, ON — SEPTEMBER 2015 – JANUARY 2016

Industry application of varied machine learning methods. Designed algorithms to perform automatic speech recognition on digital video extracted from Twitter. Implemented advanced signal processing techniques to perform acoustic modelling. Worked with a massive parallel computing architecture to process billions of data sources. Designed and built native Android & iOS apps from scratch. Culminated in award winning paper.

Video Game Developer

LUNARCH STUDIOS. WATERLOO, ON — SEPTEMBER 2014 – MAY 2015

Built a highly-robust graphics engine that supports dynamic assets loaded asynchronously. Acted as project manager to complete large-scale, internal projects. Developed a highly scalable server platform with integration between multiple software languages. Researched and implemented numerous bin-packing algorithms in order to optimize texture loading performance.

Mathematics Tutor

HUMBER COLLEGE. TORONTO, ON — JANUARY 2014 – MAY 2014

Tutored hundreds of students one-on-one in fields such as statistics, technical math, engineering, biomechanics, and business. Lead an initiative to write and produce high quality educational videos to help students with their studies. Developed a multi-platform, browser-based student sign-in system in order to collect meaningful statistics to improve effectiveness of math centre. Designed and produced graphic art to promote and develop a mathematics community.

PROGRAMMING

LANGUAGES C, C++, Python, Mathematica, Matlab, Java, Actionscript, JavaScript